

Mikhael Silveira

AI Engineer

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EDUCATION

Instituto Militar de Engenharia (IME)

Rio de Janeiro, Brazil

Bachelor in Electronic Engineering

Jan. 2023 – Dec. 2027

- Relevant Coursework: Artificial Intelligence, Machine Learning, Data Structures, Algorithms, Statistics

EXPERIENCE

Selvía

Nov. 2025 – Present

Data Science Intern — Data Engineering & AI Solutions

Remote

- Developed an LLM-based pipeline to evaluate and improve chatbot responses
- Worked on improving an existing RAG pipeline by analyzing retrieved context and identifying opportunities for improvement
- Designed an automated feedback loop where the LLM analyzed chatbot failures, identified issues in prompts or retrieved context, and proposed improvements
- Developed and maintained ELT pipelines using Python, SQL, dlt and dbt
- Integrated data from multiple sources including Pipedrive and Label Studio
- Orchestrated workflows using AWS Step Functions and GitHub Actions
- Built analytical queries and structured datasets for business decision-making
- Worked with AWS services including Athena, S3 and Glue
- Used AWS Bedrock to develop and test LLM-based approaches for automated prompt and context improvement

RoboIME

Jan. 2025 – Jan. 2026

Drone Team Lead

Rio de Janeiro, Brazil

- Developed autonomous drone systems combining computer vision, embedded systems and control
- Designed software integration between high-level autonomy and embedded flight systems
- Researched autonomous and multi-agent drone systems
- Led an 8-member engineering team developing autonomous drone systems

RoboIME

Aug. 2023 – Jan. 2025

Hardware Engineer

Rio de Janeiro, Brazil

- Developed and integrated embedded systems for autonomous drone applications
- Implemented computer vision algorithms using Python and OpenCV
- Participated in the Flying Robots Competition (CBR)

PROJECTS

LLM Search with Source Attribution | *Python, LLMs*

- Building an LLM-powered research chatbot designed to ground factual claims in verifiable sources
- Designed retrieval and response-generation workflows to improve factual reliability and traceability
- Documented architectural decisions using Architecture Decision Records (ADRs)

Drone Autonomous System | *Python, OpenCV, PyMAVLink*

- Developed computer vision and autonomous navigation solutions for drones
- Designed software integration between high-level autonomy and embedded flight systems
- Tested and deployed software in real-world autonomous flight scenarios

CEML | *Python, DSL*

- Designed an intermediate representation language for analytical representation of electronic circuits
- Developed the language as part of an AI system for solving and reasoning about electronic circuits

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, MATLAB

Cloud & Data: AWS (Athena, S3, Glue, Step Functions), MongoDB, dlt, dbt, PostgreSQL

AI: LLM Applications, AWS Bedrock, Computer Vision, OpenCV

Tools: Git, GitHub, Claude Code, Metabase